Does the Light Shift Drive Frequency Aging in the Rubidium Atomic Clock?

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***Abstract*—Frequency aging in the rubidium (Rb) vapor-cell atomic clock plays a significant role in the device's timekeeping ability. Though many researchers have speculated on the physical mechanism(s) driving the linear, deterministic frequency change (i.e., $\Delta f(t)/f_o = At$), there is little unambiguous experimental data regarding the phenomenon. Here, long-term data were used from on-orbit global positioning system (GPS) Rb clocks to examine one postulated mechanism for frequency aging (i.e., the light-shift effect). Defining the light shift of the clock's fractional frequency as $\alpha I/I_o$, where α is the light-shift coefficient, we find that temporal variations of the relative light intensity, I/I_o, cannot account for frequency aging. However, for the population of clocks considered here, we obtain the intriguing result that $\dot{\alpha}/A = 1.7 \pm 1.5$. Thus, it may be that frequency aging is driven by the light-shift effect through temporal variations of the light-shift coefficient.**

I. Introduction

Though advanced laboratory atomic standards are achieving unprecedented levels of frequency stability [1], the traditional rubidium (Rb) vapor cell and Cs beam atomic clocks remain the devices of choice for a number of applications. In no small part, this choice is due to the frequency standards' excellent long-term stability within an embodiment that is relatively compact and consumes low-to-moderate power, and is highly reliable. The issue of long-term frequency stability in these devices is of particular relevance for timekeeping because the root mean square (rms) time-error of a clock, $\delta t_{rms}(\tau)$, is given by:

$$\delta t_{rms}(\tau) \cong \frac{1}{2}A\tau^2 + k\tau\sigma_y(\tau); \tag{1}$$

here, A is the clock's deterministic linear frequency-aging rate, τ is the time interval since the clock was last synchronized and syntonized,[1] $\sigma_y(\tau)$ is the clock's Allan deviation at averaging time τ, and k is a constant on the order of unity that depends on the noise process associated with $\sigma_y(\tau)$. For Rb and Cs atomic clocks, the second term on the right-hand side of (1) increases no faster than $\tau^{3/2}$,[2] so that in the long-term linear frequency aging can play a dominant role in determining the Rb and Cs clocks' time-error.

Recent results indicate that high quality Cs atomic clocks show no evidence of fractional frequency aging above $\sim 10^{-17}$/day [2], so that the time error accrued by these devices over reasonable time intervals is solely determined by the oscillator's stochastic frequency fluctuations. Alternatively, high quality Rb atomic clocks, examined well after clock-frequency equilibration and in a benign environment,[3] tend to exhibit linear fractional frequency aging rates of 10^{-13}/day to 10^{-14}/day [3]; a set of these rates is collected in Table I for the Rb atomic clocks on board global positioning system (GPS) Block IIR satellites. Though the underlying cause(s) of frequency aging in the vapor-cell clock is not known, for a number of years the light-shift effect has been a primary suspect.

In the Rb atomic clock, the output frequency of a quartz crystal oscillator is multiplied up into the microwave regime, then locked to the Rb^{87} atom's ground state hyperfine splitting at 6834.7 MHz, ν_{hfs}, the so-called 0-0 hyperfine transition frequency [4]. In order to generate the locking signal, a population imbalance between the two hyperfine levels is created via optical pumping with an rf-discharge lamp, which additionally (and unavoidably) gives rise to a differential ac-Stark shift between the atom's ground state hyperfine levels [5], [6] (i.e., the light-shift effect). The light shift of the 0-0 hyperfine resonance frequency, ν_{LS}, is linear in the lamp's output intensity, I, and also depends on the spectral overlap between the lamp's emission lines and the atom's absorption lines. For atomic clock applications, it is typical to write $(\nu_{LS}/\nu_{hfs}) = \alpha(I/I_o)$, where I_o is the nominal light intensity entering the resonance cell and α is the light-shift coefficient, which depends on the light/atom spectral overlap [7] as well as inhomogeneous variations of light intensity [8] and spectral overlap [9] throughout the atomic-signal volume.

Manuscript received June 30, 2004; accepted November 10, 2004. This work was supported under U.S. Air Force Contract No. F040701-00-C-0009.

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[1] Just as synchronized means that the time readings of two clocks have been set to the same value, syntonized means that the frequencies of two oscillators have been set to the same value.

[2] In the Cs clock, the long-term frequency fluctuations are associated with flicker frequency noise and random-walk frequency noise, which have Allan deviations that scale like τ^0 and $\tau^{1/2}$, respectively. In the Rb clock, the long-term frequency fluctuations are associated with random-walk frequency noise.

[3] Benign here implies that the temperature, pressure, and humidity of the clock, as well as its supply voltages, are all fairly constant over the very long term. From this point of view, space is a relatively benign environment.

TABLE I

Estimated Linear Frequency Aging Rates for Six Rb Atomic Clocks Flying Onboard Block IIR GPS Satellites.[1]

GPS satellite	Estimated frequency aging rate: $\hat{A} \pm \sigma_{fit} \pm \sigma_{RW}$ $(10^{-14}/\text{day})$	$\alpha\left(\frac{\dot{I}(t)}{I_o}\right)$ $(10^{-14}/\text{day})$	$\dot{\alpha}$ $(10^{-14}/\text{day})$
SVN-41	$-3.745 \pm 0.009 \pm 0.004$	$+0.056 \pm 0.004$	-15.3 ± 7.7
SVN-43	$-2.231 \pm 0.003 \pm 0.004$		
SVN-44	$-3.484 \pm 0.006 \pm 0.004$	$+0.046 \pm 0.012$	$+8.8 \pm 14.7$
SVN-46	$-1.880 \pm 0.004 \pm 0.004$	$+0.047 \pm 0.004$	$+0.4 \pm 6.1$
SVN-51	$-3.502 \pm 0.004 \pm 0.005$	$+0.026 \pm 0.003$	-5.2 ± 20.0
SVN-54	$-3.714 \pm 0.004 \pm 0.005$	$+0.094 \pm 0.004$	$+22.4 \pm 29.3$

[1]Aging rates were determined by linear least squares fit over either 275 days or 189 days of data, and the 1-sigma uncertainties are for the fit by itself assuming only measurement noise, σ_{fit}, and for the influence of random-walk frequency noise over the fit interval, σ_{RW} [20]. To determine σ_{RW}, we assumed a worst-case, long-term Allan deviation of $\sigma_y(\tau) = 1.3 \times 10^{-16}\tau^{1/2}$ for all the clocks [21]. Note that the clock onboard SVN-43 did not exhibit frequency jumps, so it was not possible to estimate $\dot{\alpha}$ for this clock.

As noted above, researchers have often speculated on the light-shift effect as a mechanism for frequency aging:

$$A = \frac{1}{\nu_{hfs}}\frac{\delta\nu_{LS}}{\delta t} \cong \dot{\alpha} + \alpha\left(\frac{\dot{I}(t)}{I_o}\right)$$

(assuming $I(t) \cong I_o$ as demonstrated subsequently). In particular, very slow changes in the lamp's light intensity have been observed, and it has been hypothesized that these must affect the clock's frequency through the light-shift effect, thereby giving rise to frequency aging. This tidy picture was confounded in the mid-1980s, however, when Volk and Frueholz [10] demonstrated that

$$\left|\alpha\left(\frac{\dot{I}(t)}{I_o}\right)\right| \ll |A|$$

for one high quality Rb clock. Of course, it could be that Volk and Frueholz's sample-of-one result corresponded to the exception rather than the rule; and, as mentioned by Camparo [11], there is always the possibility that $\dot{\alpha}$ is nonzero. Consequently, given the present empirical situation, it must be admitted that the light-shift effect remains a viable candidate for the frequency aging mechanism, though the situation is certainly confused. In the following, we attempt to clarify the issue by examining frequency aging and the light-shift effect for a set of high quality Rb atomic clocks.

II. Investigations

Recently, we investigated on-orbit frequency equilibration for a number of GPS Block IIR Rb clocks; an example of the data is presented in Fig. 1 [12]. In five out of six cases, we found that the satellite clocks displayed relatively sudden changes in lamp intensity that coincided with sudden changes in clock fractional frequency. Two examples of this phenomenon are shown in Fig. 2, in which we have removed the deterministic linear frequency aging and frequency equilibration from the data for a cleaner analysis of the fractional-frequency/light-intensity jumps. Notice that, in one case, the data correspond to clock operation a little over 6 months after the clock was turned on, and in the other case about 2 years after turn on. Thus, at the time of these events, the clock's lamp, filter cell, and resonance cell had achieved their nominal operating conditions. (Notice also that, to within a few percent, $I(t)/I_o$ is unity.) Consequently, the events shown in Figs. 1 and 2 cannot be associated with the clock's warm-up. Therefore, we used the amplitude of these events to estimate $\alpha(t)$ at various times during the clock's operating history. These estimates for several of the satellite clocks are shown in Fig. 3, and the linear fit values of $\dot{\alpha}$ for all five of the satellite clocks are collected in Table I. Also shown in Table I are estimates of $\alpha\left(\dot{I}(t)/I_0\right)$ for the satellite clocks; these were obtained by using an average value of $\alpha(t)$ for each

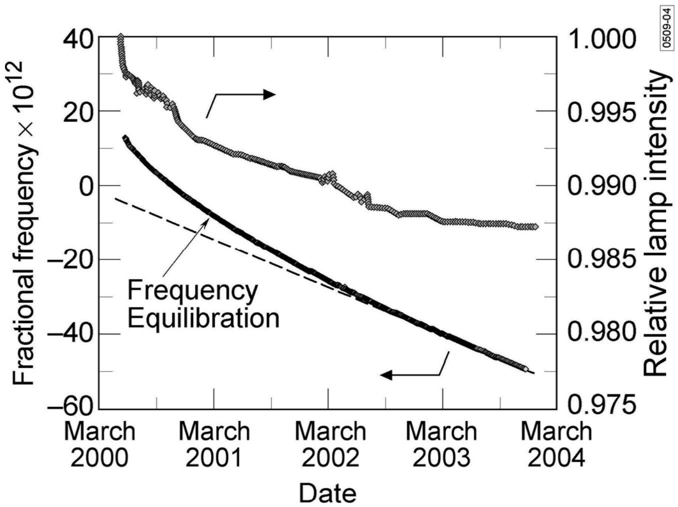

Fig. 1. History of the fractional frequency and lamp intensity for SVN-51. As the figure shows, there is a slow equilibration period followed by linear frequency aging. The coefficients of linear frequency aging, A, and relative lamp intensity, $\dot{I}(t)/I_o$, were determined from the data between June 1, 2003, and December 6, 2003, for this satellite clock. The values for the other satellite clocks also were determined after the clocks had achieved equilibrium.

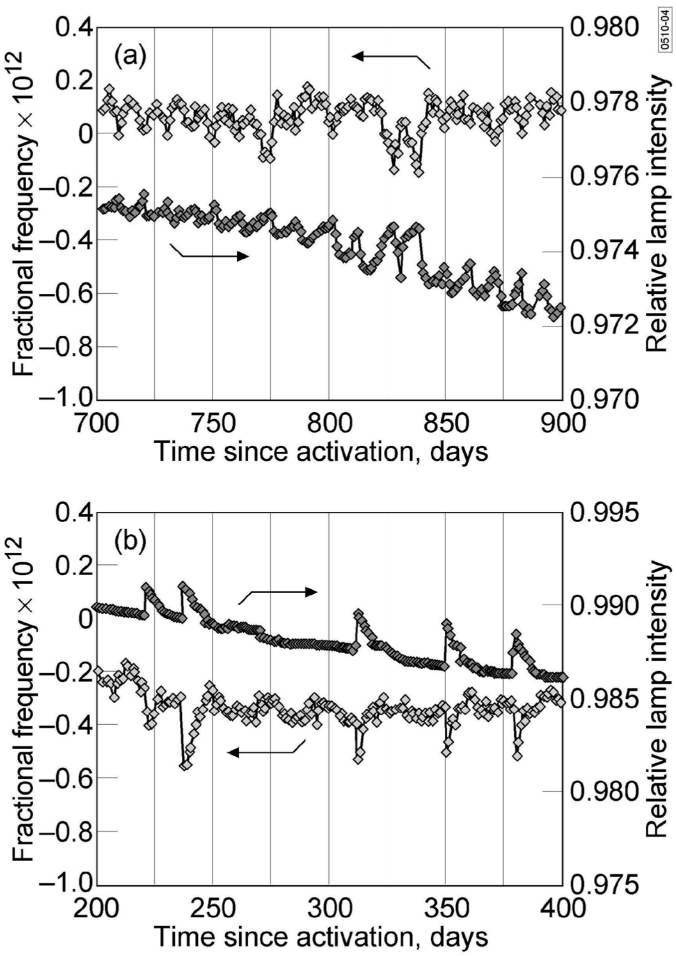

Fig. 2. This figure shows jumps in the lamp intensity that coincide with jumps in the clock's frequency: (a) Rb clock on board GPS satellite SVN-46, and (b) Rb clock on board GPS satellite SVN-54. We attribute this behavior to the light-shift effect as discussed in the text.

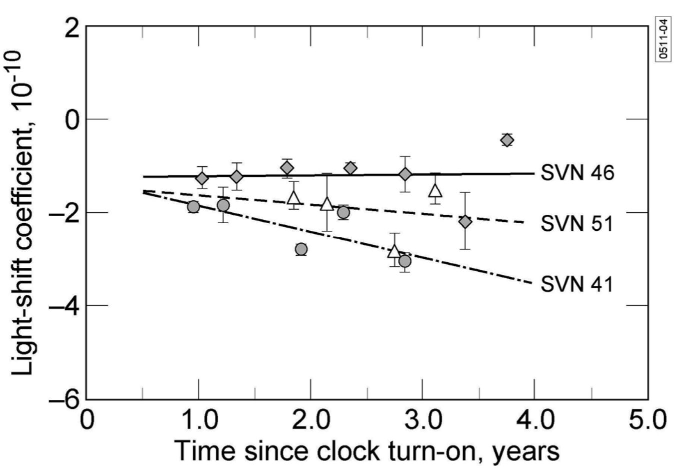

Fig. 3. Light-shift coefficient as a function of time since clock activation on orbit. Circles correspond to the clock onboard SVN 41, triangles correspond to the clock onboard SVN 51, and diamonds correspond to the clock onboard SVN 46. Data points correspond to the periods in the data record in which an unambiguous determination of the light-shift coefficient could be made based on correlated frequency and light-intensity jumps.

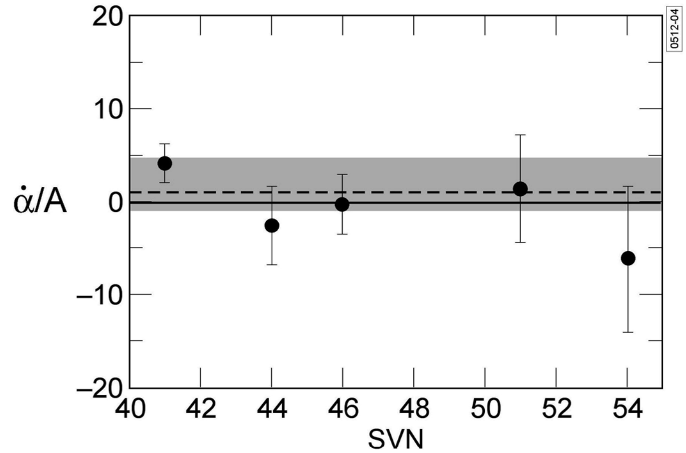

Fig. 4. For each of the space vehicles, the ratio of the estimated derivative of the light-shift coefficient, $\dot{\alpha}$, and the linear frequency-aging rate, A, is given. The dashed line corresponds to $\dot{\alpha}/A = 1$. The gray region represents the 95% confidence interval for the weighted mean of all the space-vehicle ratios.

clock, and using data well past equilibration to estimate $\left(\dot{I}(t)/I_o\right)$.

III. Results

As evidenced by the data collected in Table I, our results are consistent with the Volk-Frueholz finding. The lamps' intensity changes are too small to account for frequency aging via the light-shift effect; moreover, they are of the wrong sign. However, regarding a possible frequency-aging role for the light-shift coefficient, our results are less definitive. In particular, Fig. 4 shows the various clock values of $\dot{\alpha}/A$. Although the uncertainty of the $\dot{\alpha}/A$ estimates is large for the individual satellite clocks, it is possible to construct a weighted mean value, $\langle\dot{\alpha}/A\rangle$, for the sample [13] because all of these clocks are identically constructed and operate under identical conditions. In this way we obtain the intriguing result $\langle\dot{\alpha}/A\rangle = 1.7 \pm 1.5$. We must note, however, that even with this weighted mean value we cannot reject the hypothesis $\langle\dot{\alpha}/A\rangle = 0$ with anything better than a 0.87 probability. (In Fig. 4 the shaded region corresponds to the 95% confidence interval for the value of $\langle\dot{\alpha}/A\rangle$.)

Though more work is clearly required on what it will be convenient to refer to as the "$\dot{\alpha}$-hypothesis," it is reasonable in light of our findings to speculate on how a nonzero derivative of the light-shift coefficient might arise. One likely explanation for $\dot{\alpha} \neq 0$ posits a slow change in spectral overlap between the lamp's emission lines and the atom's absorption lines. Because the lamp's emission spectrum is a sensitive function of alkali density, it might be that slow changes in the lamp's operating temperature change the spectral overlap [14], [15]. Alternatively, alkali reaction with the lamp's glass envelope could lead to the production of volatile and reactive species that coat the liquid pool of metal within the lamp, thereby reducing the alkali vapor pressure [16], [17]. Similarly, outgassing of reactive species (e.g., O_2 or water) could lead to a reduction of alkali density [18]. Filtering of the lamp lines, either by a separate filter cell or within the clock's resonance cell, also affects the spectral overlap and, therefore, changes in filtering action could lead to a non-zero value of $\dot{\alpha}$. These changes in filtering action could again be driven by temperature variations in the filter and/or resonance cell. Given the inhomogeneous nature of the light shift in the vapor-cell atomic clock [8], any variations in the resonance-cell region in which the atomic signal is generated [19] could give rise to an affect that appears as a non-zero $\dot{\alpha}$.

IV. Summary

We have used long-term data from on-orbit GPS Rb clocks to study the light-shift effect as a possible mechanism for frequency aging. Consistent with the sample-of-one study of Volk and Frueholz [10] in the mid-1980s, temporal variations of the lamp's light intensity cannot give rise to frequency aging through the light-shift effect for these clocks. Regarding possible variations in the light-shift coefficient as a source of frequency aging, our data is inconclusive, though highly suggestive that $\dot{\alpha}/A = 1$. In this regard, we note that ours are the first estimates of $\dot{\alpha}$, and that no other empirical work regarding such a variation exists in the literature. Based on the results presented here, further investigation of the $\dot{\alpha}$-hypothesis seems warranted.

Acknowledgments

The author would like to acknowledge the assistance of Mr. M. Epstein and Mr. T. Dass for their help in obtaining the on-orbit GPS data.

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